

# Mikael Rinne

Founding Engineer · Rust, C#, TypeScript

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## PROFILE

Founding engineer who builds developer tools and real-time systems in **Rust, C# and TypeScript**. Top contributor to **Ivy Framework** (435 GitHub stars) and lead engineer of **Ivy Tendril**, a development environment for teams where AI agents write most of the code. Founder of **SpaceCorps**, which ships a space MMO on its own Rust engine and maintains 20+ open-source Rust CLIs for AI agents. Started in computational physics and machine learning, with a first-author IEEE paper written with CERN collaborators.

## EXPERIENCE

### Ivy Interactive · 1st Founding Engineer

2025 – present · Stockholm

*A .NET AI startup: a pure-C# full-stack framework for internal tools with LLM code generation, and Tendril, an agentic development environment.*

- Top contributor to **Ivy Framework** (C#/.NET, 435 stars): **3,586 commits and 1,148 PRs**, 38% of the repository's history.
- Lead engineer of **Ivy Tendril** since its first commit in April 2026 (201 stars; macOS, Windows, Linux): **2,128 commits and 652 PRs**, 48% of the repository. Agents run in parallel in isolated git worktrees and changes are reviewed and merged in-app. Now leading the V2 rewrite as a TypeScript and Rust monorepo.
- Wrote much of the Rust layer: Rusty-Framework (pure-Rust port of Ivy), Ivy.NativeJsonDiff (RFC 6902 diffing over Rust FFI for reactive UI updates), Ivy.Docs.Compiler, Tendril-Service (headless agent orchestration service and CLI), and co-built Rustino (native webview windows, a Photino.NET replacement).
- Own product engineering, AI modelling and planning, and lead a team of developers. Across the company's GitHub organisation: **2,700+ PRs authored and 600+ reviewed**.
- Speaker at the Agentics Foundation "Harness Engineering" meetup (IBM Innovation Studio) as maintainer of Ivy Tendril; author of the talk "Three things that nearly killed our software factory".

### SpaceCorps Technology OÜ · Founder & CEO / CTO

2023 – present · Remote, Europe

- Founded the company around an open-source 3D MMO; recruited and lead an international team of six (designers and full-stack developers). Built the original WebGL prototype, SpaceCorps 2 (TypeScript, three.js), myself.
- Built **Space3d**, an in-house game engine in Rust on wgpu/WebGPU targeting native, web and mobile, and **SpaceCorps 2027** on top of it: a space MMO with public builds for Windows, macOS and Linux (latest build v0.4.5).
- Published **20+ MIT-licensed Rust CLIs** that give AI agents safe access to real services (Gmail that can only draft, Cloudflare, Firecrawl, Exa, Sliplane, Namecheap and more): structured output, stable exit codes, OS-keychain credentials and prompt-injection guards. Co-built with Niels Bosma.
- Open-sourced open-apple-models and open-android-models (Swift, Kotlin; pre-release): tool calling for on-device LLMs (Apple Foundation Models, Gemini Nano) to drive game NPCs.

### Scania · Solutions Architect / Senior Full-stack Engineer

2024 – 2025 · Stockholm

- Modernised the internal platform used to deliver trucks, improving the user experience and supporting sales.
- Proposed and implemented new workflows and algorithms during the platform's migration to AWS.

### Nixor EE AS · Full-stack Software Engineer

2023 – 2024 · Tallinn

- Built and shipped SaaS and POS improvements for Coop, Stockmann Group and Selver, using custom Odoo modules and payment integrations.
- Found and fixed critical transaction bugs in production retail systems across Estonia.

### University of Tartu, Institute of Technology · Software Engineer & Researcher

2022 – 2023 · Tartu

- Built scientific web software for molecular dynamics and quantum emission calculations with CERN collaborators (GETELEC, 174 commits), bridging theoretical and experimental physics.
- First author of "An Online Tool for Thermal-Field Emission Calculations from Metal and Semiconducting Emitters", IEEE IVNC 2023.

Private tutor, natural sciences

2019 – 2024 · Tallinn, remote

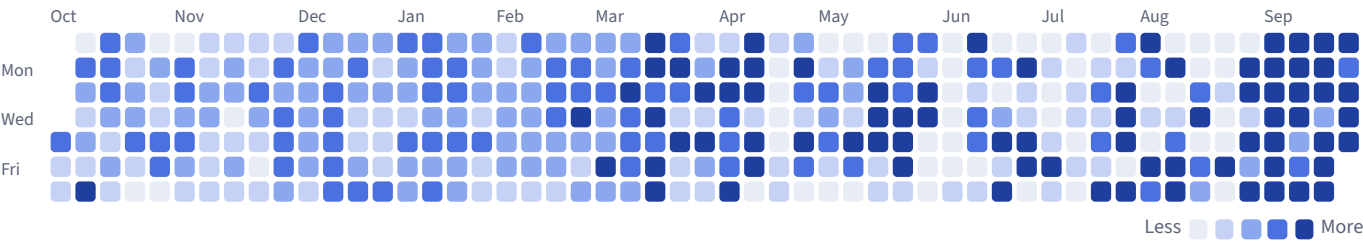
- Taught natural sciences one-on-one, in groups and remotely, helping students improve their academic results. Mentoring is still central to how I run a team.

OPEN SOURCE

|                                                                                                   |                                                                                                                           |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Ivy Framework</b> · C#/.NET, 435 stars<br>github.com/Ivy-Interactive/Ivy-Framework             | <b>open-apply</b> · Rust CLI for AI-assisted job search and application tracking, MIT<br>github.com/SpaceCorps/open-apply |
| <b>Ivy Tendril</b> · agentic dev environment, 201 stars<br>github.com/Ivy-Interactive/Ivy-Tendril | <b>open-apple-models</b> · on-device LLM tool calling, Swift, MIT<br>github.com/SpaceCorps/open-apple-models              |
| <b>SpaceCorps CLI fleet</b> · 20+ Rust CLIs for agents, MIT<br>github.com/SpaceCorps              | <b>Rusty-Framework</b> · full-stack web framework in pure Rust<br>github.com/Ivy-Interactive/Rusty-Framework              |

GITHUB ACTIVITY, LAST 365 DAYS

21,370 contributions in the last 365 days · 312 active days · longest streak 140 days



EDUCATION & HONORS

|                                                                                                                                                                                                                                                                           |             |                                                            |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|------|
| <b>SALT School of Applied Technology</b><br>Full-stack C# developer programme: TDD, mob programming, ASP.NET, Blazor, EF Core, React.                                                                                                                                     | 2024        | <b>Bronze medal, 51st International Chemistry Olympiad</b> | 2019 |
| <b>University of Tartu, Chemistry</b><br>Studied chemistry with a heavy computational and data science focus: ML models for molecular dynamics (Gaussian approximations, Monte Carlo, neural networks, gradient boosting) and simulation in COMSOL, Modelica and Fortran. | 2019 – 2024 | <b>1st place, Estonian National Chemistry Olympiad</b>     | 2018 |
|                                                                                                                                                                                                                                                                           |             | <b>Google: Foundations of Data Science, Python</b>         | 2024 |

SKILLS

|                                |                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>               | Rust, C#, TypeScript, Python, SQL; also Swift, Kotlin, C/C++, Java, Fortran, WGSL                                             |
| <b>Backend &amp; cloud</b>     | .NET 10, ASP.NET Core, EF Core, SignalR, axum, Tokio, PostgreSQL, Docker, Kubernetes, AWS, Azure, GitHub Actions              |
| <b>Frontend &amp; graphics</b> | React, Vite, Tailwind, TanStack, three.js/WebGL, wgpu/WebGPU                                                                  |
| <b>AI engineering</b>          | Agent orchestration, tool calling, MCP, Anthropic and OpenAI APIs, RAG and vector search, on-device models, evals             |
| <b>Also used</b>               | Angular, Node.js, Dapper, Odoo, SQL Server, MySQL, xUnit, Vitest, Jest, Jira, Confluence, COMSOL, Modelica, MATLAB, Assembler |
| <b>Practice</b>                | TDD, SOLID, agile, CI/CD, code review, mob programming, technical leadership, hiring and mentoring                            |
| <b>Human languages</b>         | English (native), Estonian (fluent), Russian (fluent)                                                                         |